GEOGRAPHY

PHYSICAL GEOGRAPHY

GEOMORPHOLOGY

MODULE - 33

FLUVIAL LANDFORMS

- DEPOSITIONAL LANDFORMS
- DRAINAGE PATTERNS

FLUVIAL LANDFORMS

DEPOSITIONAL LANDFORMS

2 DELTAS

1. Deltas are like alluvial fans but develop at a different location.
2. The load carried by the rivers is dumped and spread into the sea.
3. If this load is not carried away far into the sea or distributed along the coast, it spreads and accumulates as a low cone.
4. Unlike in alluvial fans, the deposits making up deltas are very well sorted with clear stratification.

FLUVIAL LANDFORMS

DEPOSITIONAL LANDFORMS

2 DELTAS

A Satellite view of part of Krishna River delta, Andhra Pradesh

5 The coarsest materials settle out first and the finer fractions like silts and clays are carried out into the sea.

6 As the delta grows, the river distributaries continue to increase in length and delta continues to build up into the sea.

FLUVIAL LANDFORMS

DEPOSITIONAL LANDFORMS

3 FLOODPLAINS

1. Deposition develops a floodplain just as erosion makes valleys. Floodplain is a major landform of river deposition.

2. Large sized materials are deposited first when stream channel breaks into a gentle slope.

3. Thus, normally, fine sized materials like sand, silt and clay are carried by relatively slow moving waters in gentler channels usually found in the plains.

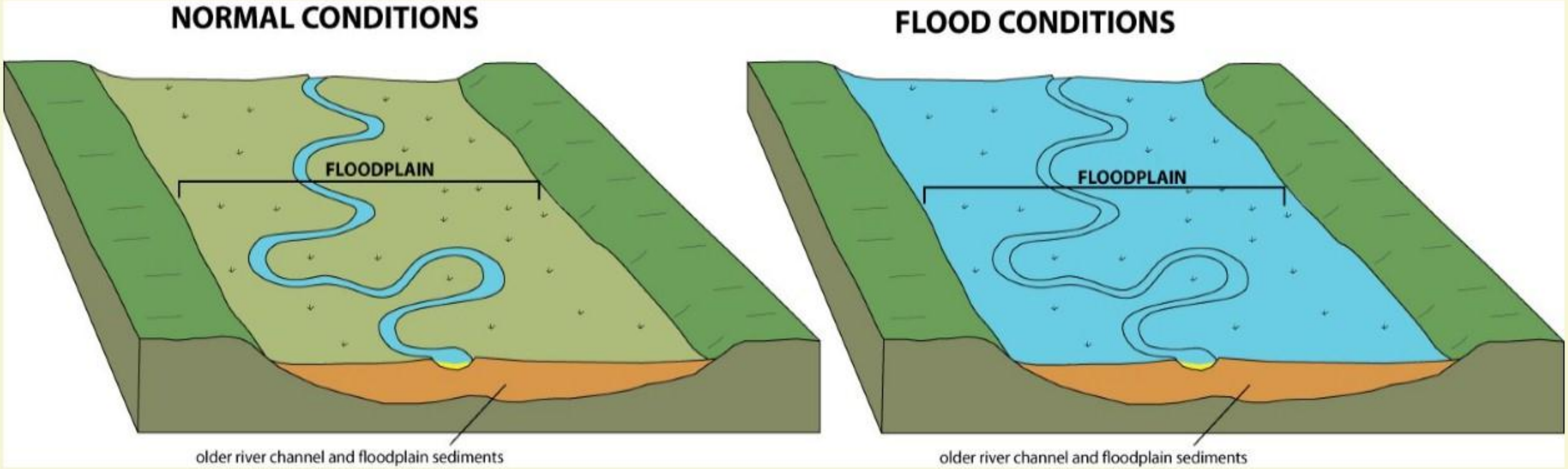
4. It gets deposited over the bed and the waters spill over the banks during flooding. A river bed made of river deposits is the active floodplain. The floodplain above the bank is inactive floodplain.

5. Inactive floodplain above the banks basically contain two types of deposits flood deposits and channel deposits.

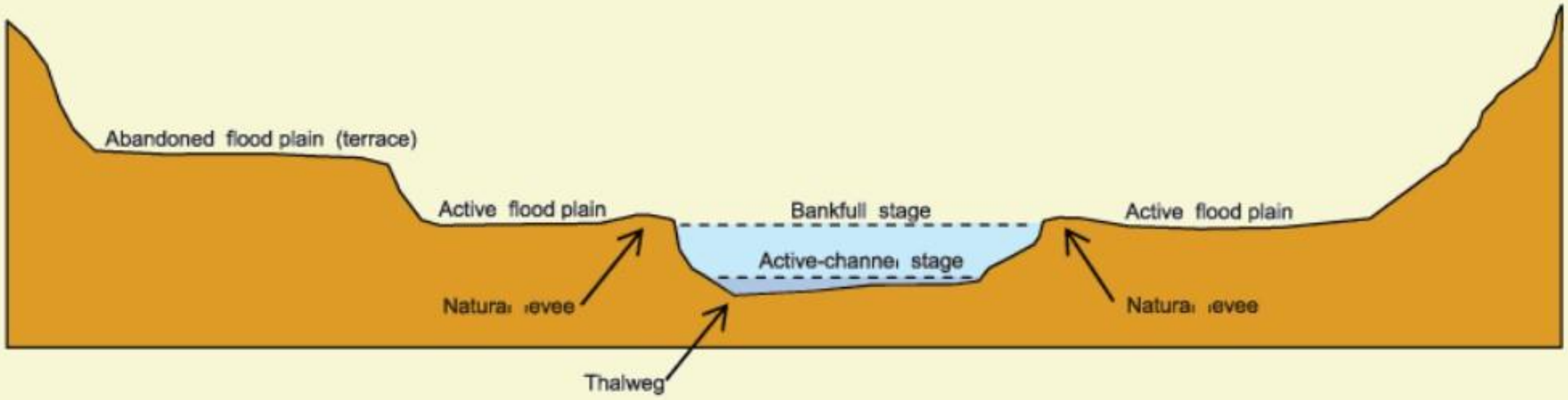
FLUVIAL LANDFORMS

DEPOSITIONAL LANDFORMS

3 FLOODPLAINS



ACTIVE AND INACTIVE FLOODPLAINS



FLUVIAL LANDFORMS

DEPOSITIONAL LANDFORMS

3 FLOODPLAINS

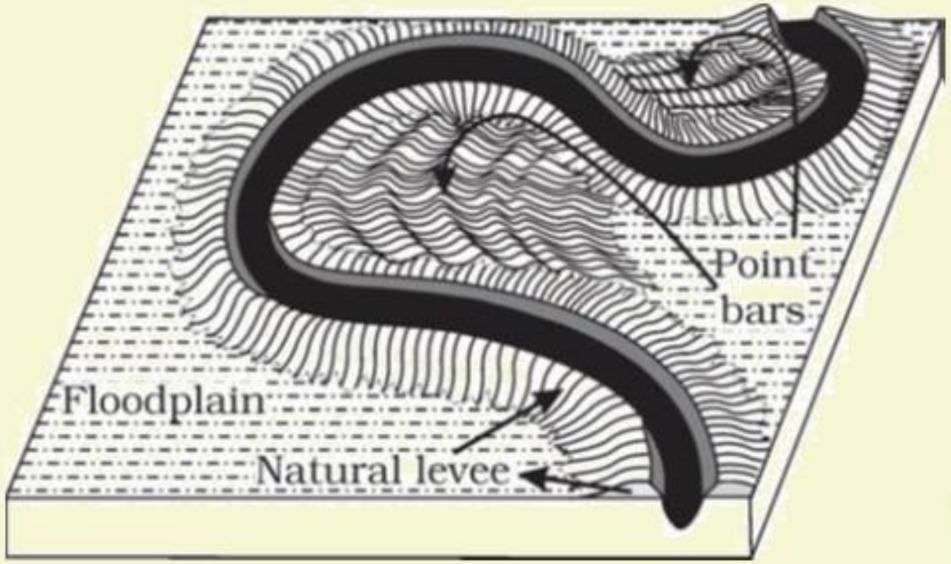
6 In plains, channels shift laterally and change their courses occasionally leaving cut-off courses which get filled up gradually.

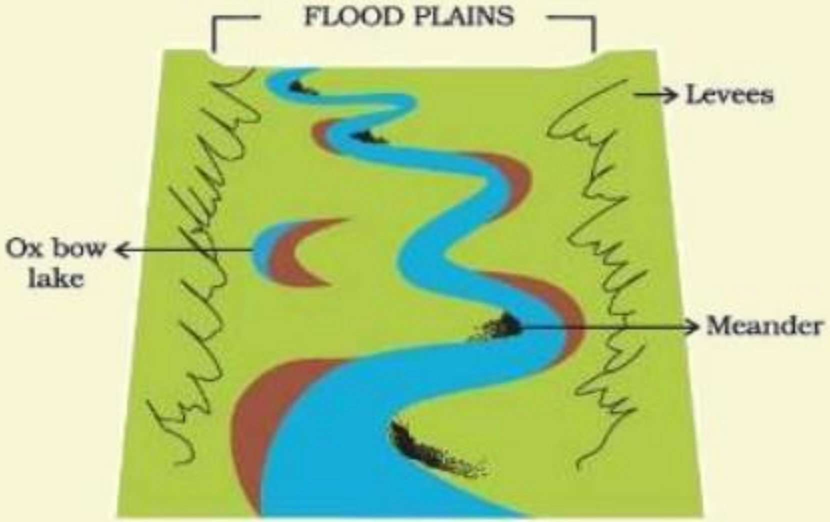
7 Such areas over flood plains built up by abandoned or cut-off channels contains coarse deposits.

8 The flood plains in a delta are called delta plains.

DEPOSITIONAL LANDFORMS

4 NATURAL LEVEES





1. Natural levees and point bars are some of the important landforms found associated with floodplains.

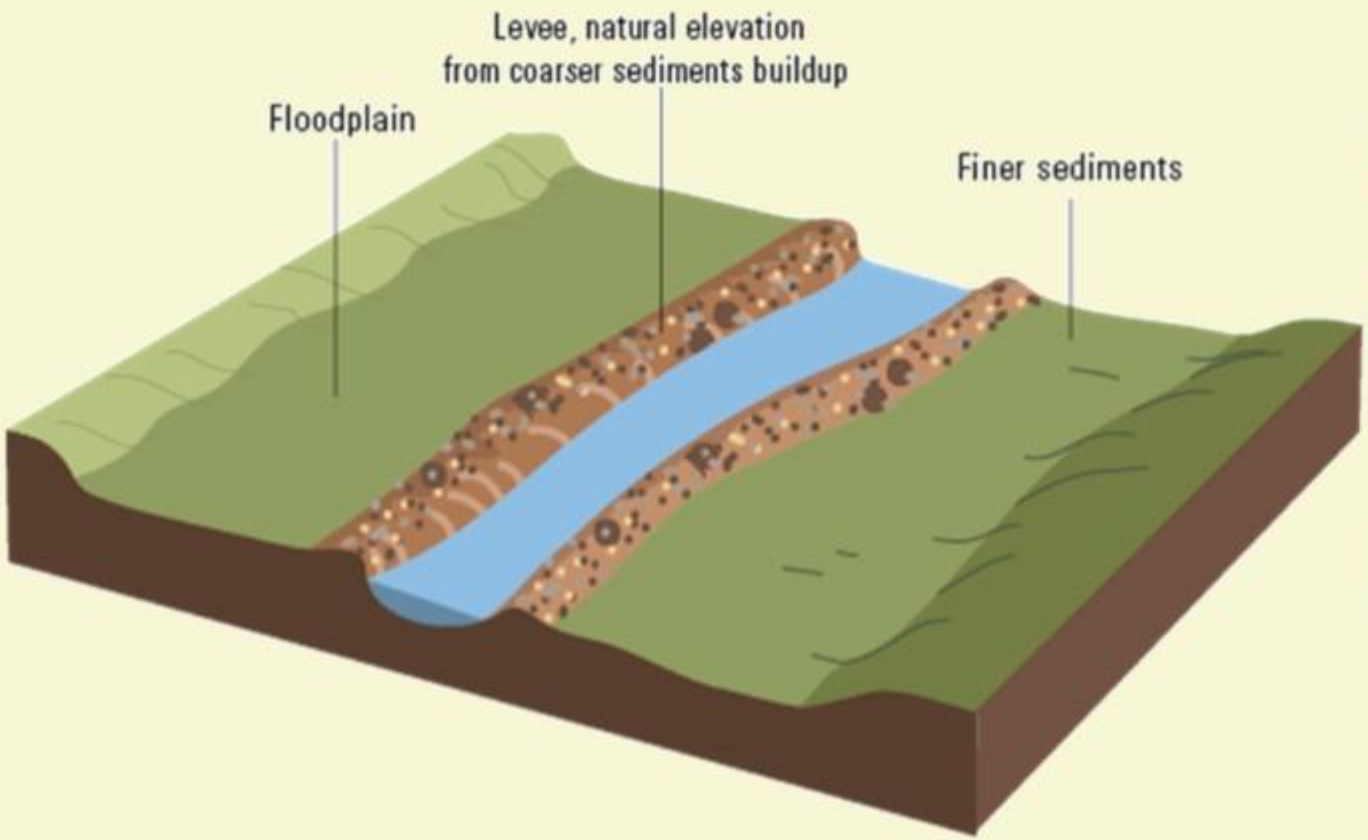
2. Natural levees are found along the banks of large rivers. They are low, linear and parallel ridges of coarse deposits along the banks of rivers, quite often cut into individual mounds.

3. During flooding as the water spills over the bank, large sized materials get dumped in the immediate vicinity of the bank as ridges.

FLUVIAL LANDFORMS

DEPOSITIONAL LANDFORMS

4 NATURAL LEVEES

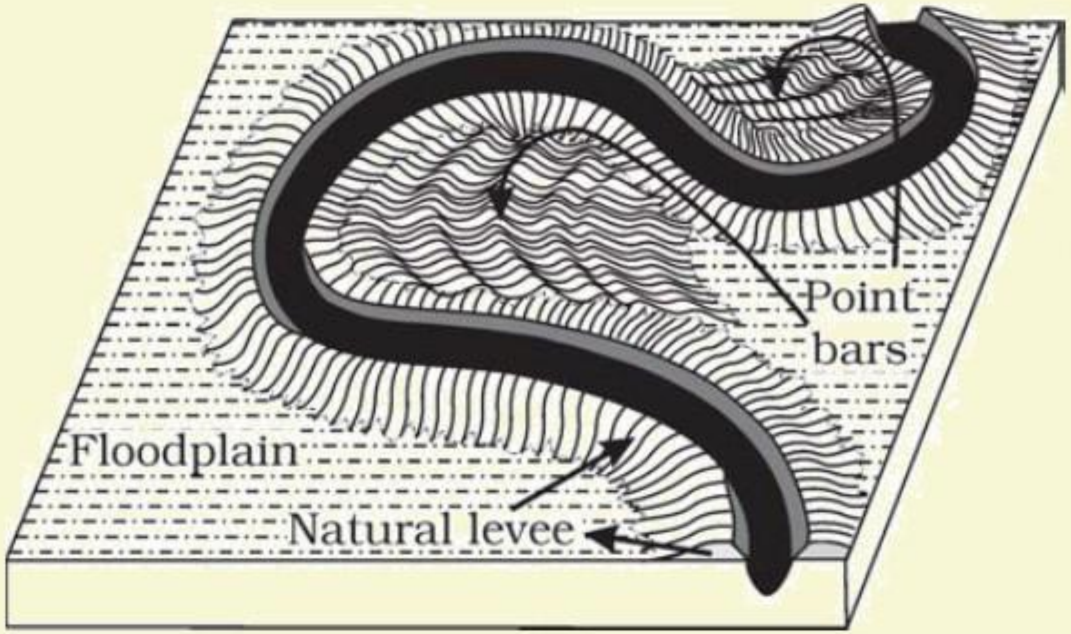


4. They are high near the banks and slope gently away from the river.
5. The levee deposits are coarser than the deposits spread by flood waters away from the river.
6. When rivers shift laterally, a series of natural levees can form.

FLUVIAL LANDFORMS

DEPOSITIONAL LANDFORMS

5 POINT BARS

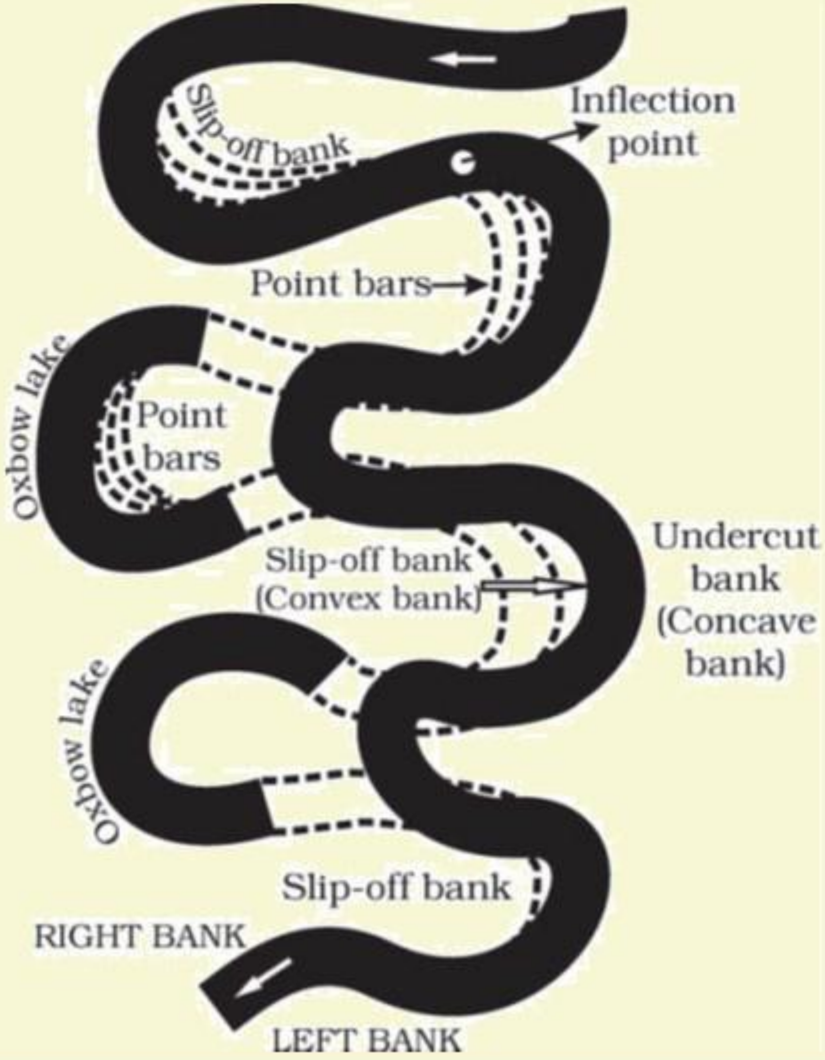


1. Point bars are also known as meander bars. They are found on the convex side of meanders of large rivers and are sediments deposited in a linear fashion by flowing waters along the bank.
2. They are almost uniform in profile and width and contain mixed sizes of sediments.
3. Rivers build a series of such point bars depending upon the water flow and supply of sediment.
4. As the rivers build the point bars on the convex side, the bank on the concave side will erode actively.

FLUVIAL LANDFORMS

DEPOSITIONAL LANDFORMS

6 MEANDERS



1. In Large flood and delta plains, rivers rarely flow in straight courses. Loop-like channel patterns called Meanders develop over flood and delta plains.

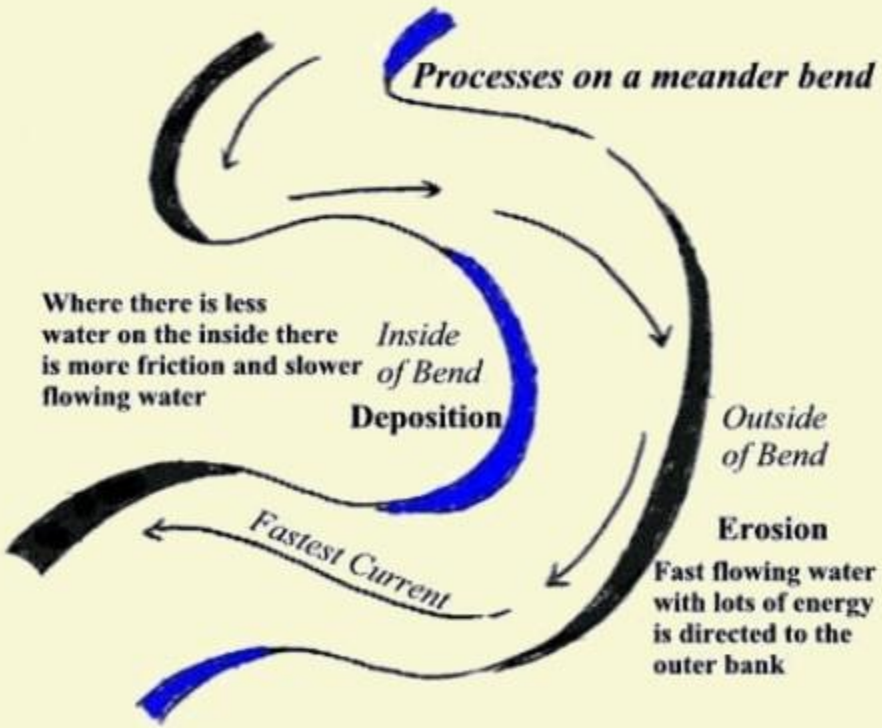
2. Meander is not a landform but is only a type of channel pattern. This is because of
   i. Propensity of water flowing over very gentle gradients to work laterally on the banks.
   ii. Unconsolidated nature of alluvial deposits making up the banks with many irregularities which can be used by water exerting pressure laterally.
   iii. Coriolis force acting on the fluid water deflecting it like it deflects the wind.

FLUVIAL LANDFORMS

DEPOSITIONAL LANDFORMS

6 MEANDERS

Meander Formation



3. When the gradient of the channel becomes extremely low, water flows leisurely and starts working laterally.

4. Slight irregularities along the banks slowly get transformed into a small curvature in the banks.

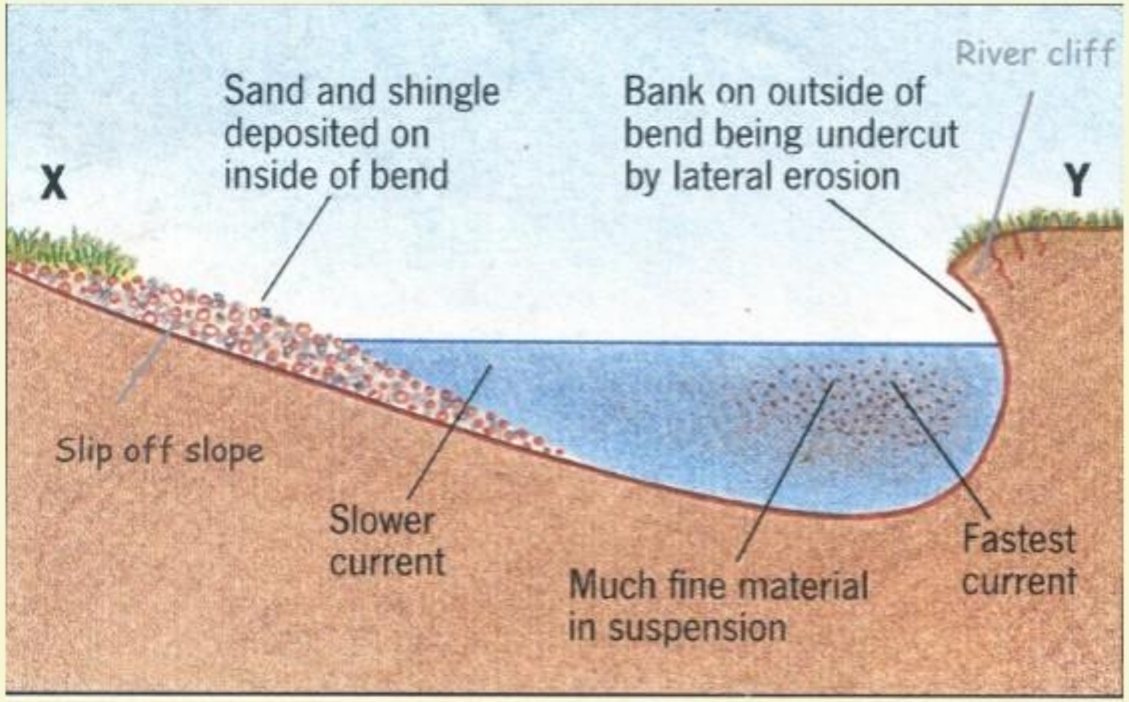
5. The curvature deepens due to deposition on the inside of the curve and erosion along the bank on the outside.

6. If there is no deposition and no erosion or undercutting, the tendency to meander is reduced.

FLUVIAL LANDFORMS

DEPOSITIONAL LANDFORMS

6 MEANDERS



7. Normally, in meanders of large rivers, there is active deposition along the convex bank and undercutting along the concave bank.

8. The concave bank is known as cut-off bank which shows up as a steep scarp and the convex bank presents a long, gentle profile and is known as slip-off bank

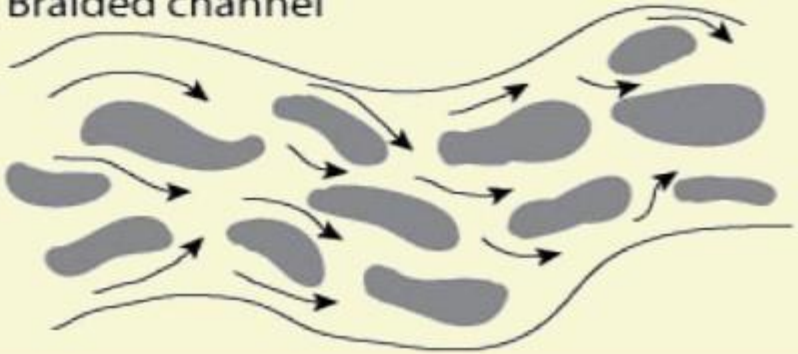
9. As meanders grow into deep loops, the same may get cut-off due to erosion at the inflection points and are left as ox-bow lakes.

FLUVIAL LANDFORMS

DEPOSITIONAL LANDFORMS

7 BRAIDED CHANNELS

Braided channel



1. When rivers carry coarse material, there can be selective deposition of coarser materials causing formation of a central bar which diverts the flow towards the banks and this flow increases lateral erosion on the banks.

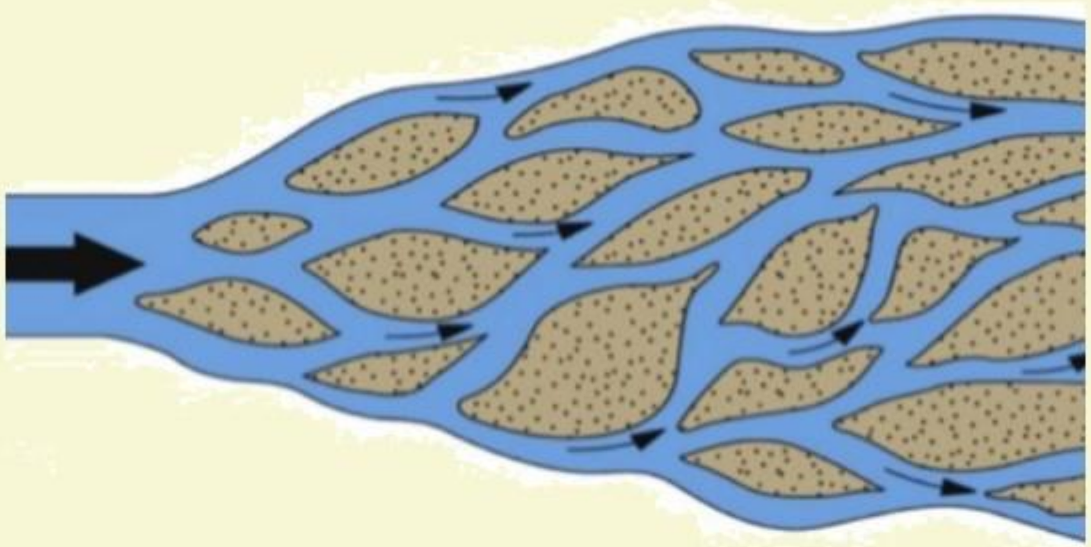
2. As the valley widens, the water column is reduced and more and more materials get deposited as islands and lateral bars developing a number of separate channels of water flow.

3. Deposition and lateral erosion of banks are essential for the formation of braided pattern.

FLUVIAL LANDFORMS

DEPOSITIONAL LANDFORMS

7 BRAIDED CHANNELS



Braiding occurs when the stream discharge is insufficient to transport the available load.

4 Or, alternatively, when discharge is less and load is more in the valley, channel bars and islands of sand, gravel and pebbles develop on the floor of the channel and the water flow is divided into multiple threads.

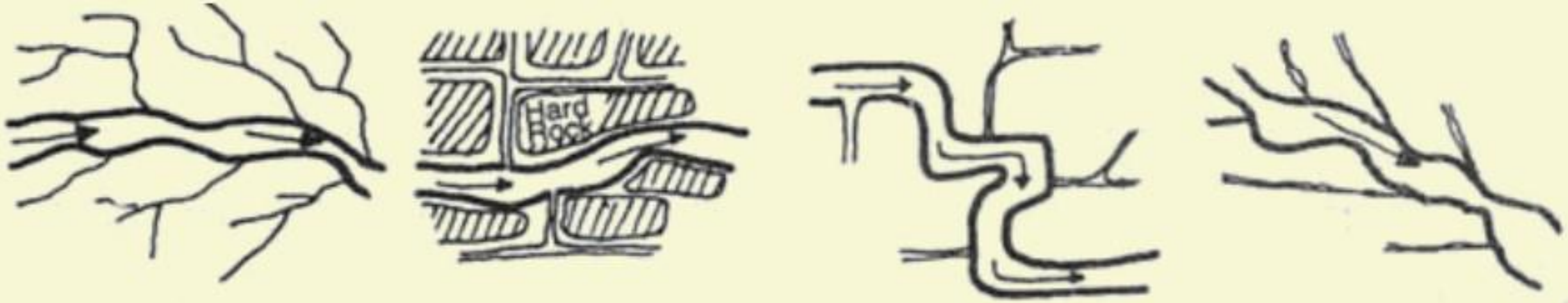
5 These thread-like streams of water re-join and subdivide repeatedly to give a typical braided pattern.

FLUVIAL LANDFORMS

DRAINAGE PATTERNS

1 The typical shape of a river course as it completes its erosional cycle is referred to as the drainage pattern of a stream.

2 A drainage pattern reflects the structure of basal rocks, resistance and strength, cracks or joints and tectonic irregularity.

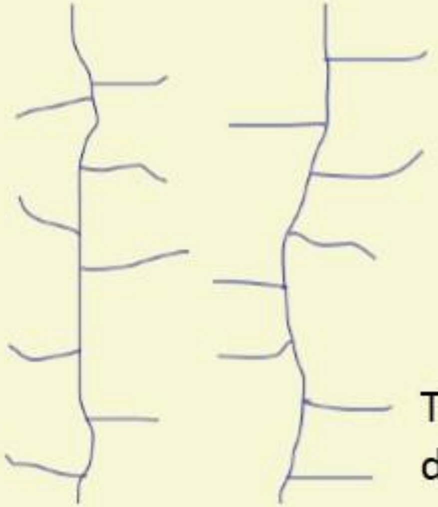


(i) Dendritic (ii) Trellis (iii) Rectangular (iv) Angular

(v) Parallel (vi) Radial (vii) Annular (iv) Centripetal

FLUVIAL LANDFORMS

DRAINAGE PATTERNS



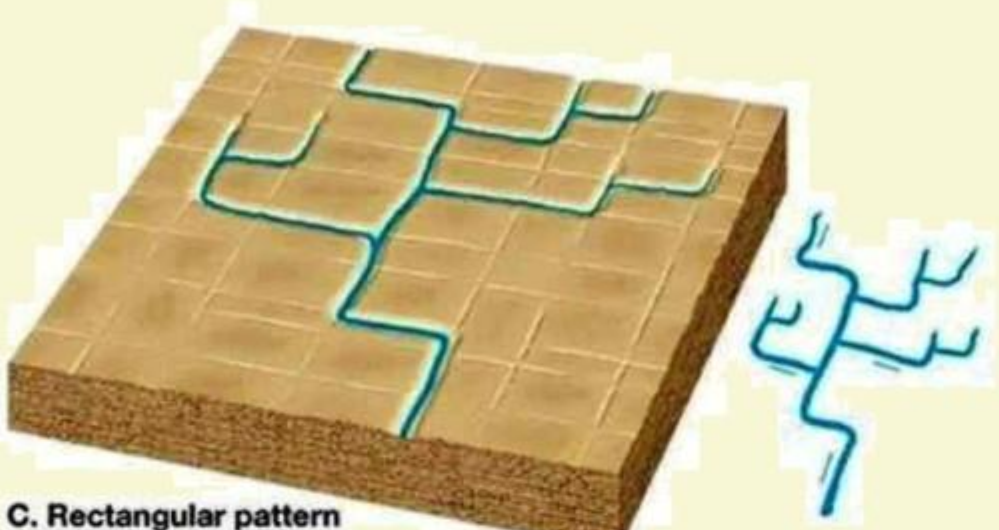
Trellis drainage

TRELLIS

1. When the primary tributaries of rivers flow parallel to each other and secondary tributaries join them at right angles, the pattern is known as 'trellis'. A trellis drainage pattern develops where hard and soft rocks lie parallel to each other.
2. Examples: Rivers in upper part of Himalayas; folded Singhbhum mountains

FLUVIAL LANDFORMS

DRAINAGE PATTERNS



C. Rectangular pattern develops on highly jointed bedrock

RECTANGULAR

1. The main stream bends at right angles and the tributaries join at right angles creating rectangular patterns.
2. It differs from trellis pattern drainage, since it is more irregular and its tributary streams are not as long or as parallel as in trellis drainage.
3. This pattern has a subsequent origin (subsequent drainage – we shall study in Indian drainage systems).
   Example: Colorado river (USA), streams in Vindhyan mountains.

FLUVIAL LANDFORMS

DRAINAGE PATTERNS

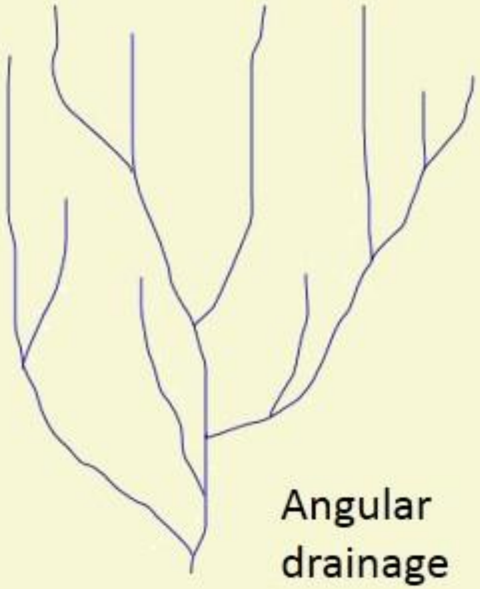
DENDRITIC OR PINNATE

1. This is an irregular tree branch shaped
2. Examples: Rivers of Northern Plains

A. Dendritic pattern develops on relatively uniform bedrock

ANGULAR

1. The tributaries join the main stream at acute angles.
2. This pattern is common in Himalayan foothill regions.



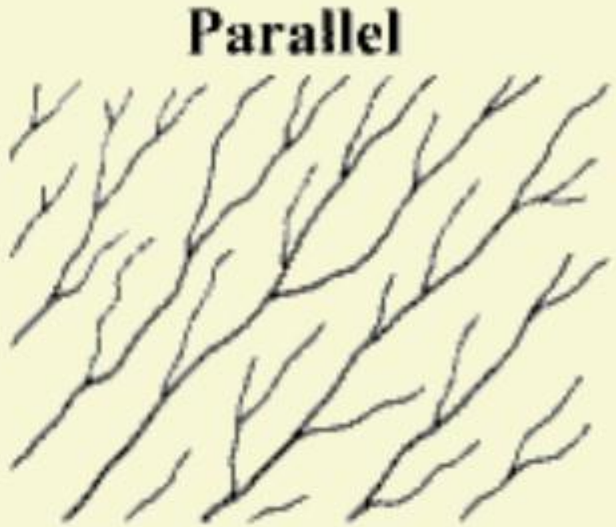
Angular drainage

FLUVIAL LANDFORMS

DRAINAGE PATTERNS

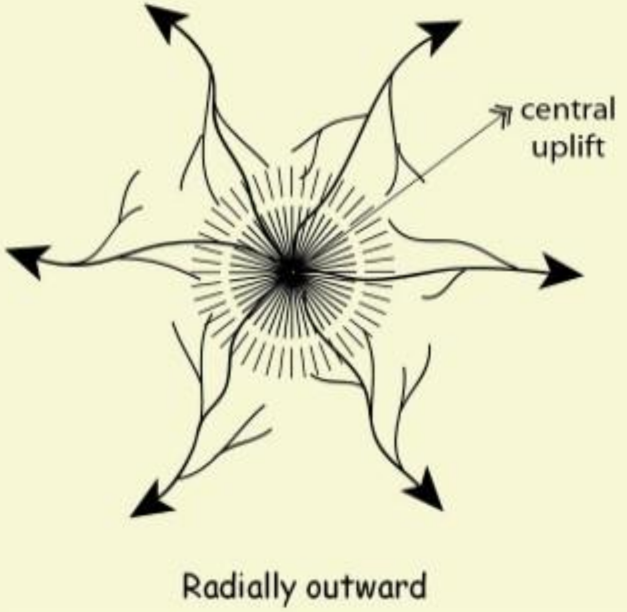
PARALLEL

1. The tributaries seem to be running parallel to each other in a uniformly sloping region.
2. Example: rivers of Western Ghats draining into Arabian Sea.



RADIAL

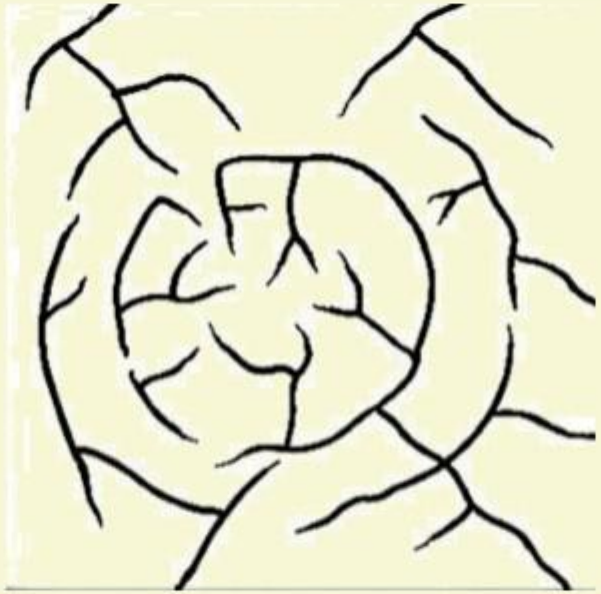
1. The tributaries from a summit follow the slope downwards and drain down in all directions.
2. Examples: streams of Amarakantak plateau



DRAINAGE PATTERNS

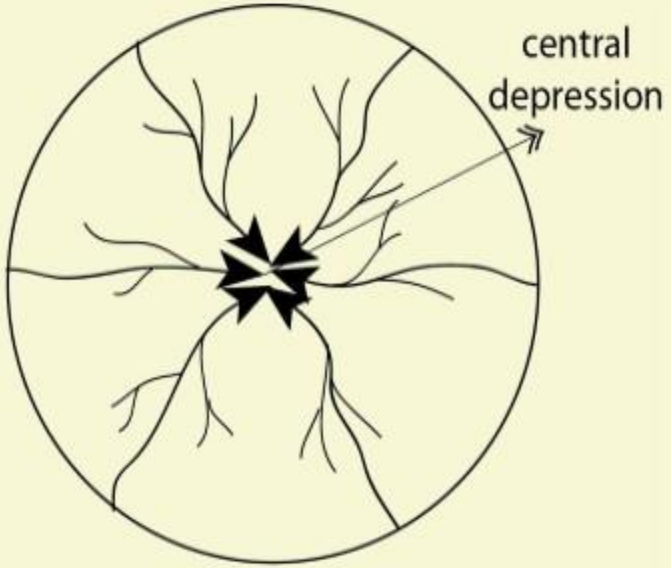
ANNULAR

1. It represents that part of a drainage pattern in which the subsequent streams follow the curving or arcuate courses before joining the consequent stream.
2. These results from a partial adaptation to an underground circular batholith structure. E.g.: Black Hill streams of South Dakota.



CENTRIPETAL

1. Into a low lying basin the streams converge from all sides.
2. Examples: streams draining into Loktak lake, Manipur.



Radially inward

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